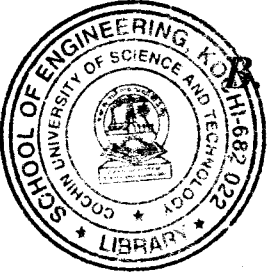


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B.Tech. Degree VII Semester Examination November 2018

ME 15-1703 MACHINE DESIGN II (2015 Scheme)

(Use of Design Data Handbook is permitted. Data not given may be suitably assumed.
All such assumptions must be clearly stated.)

Time: 3 Hours

Maximum Marks: 60

PART A (Answer ALL questions)

(10 × 2 = 20)

- I. (a) Differentiate between single plate and multi plate clutches with application.
- (b) Explain the effect of centrifugal tension in belt on power transmission.
- (c) Explain the working of band brakes.
- (d) What are the points to be considered while designing a sliding mesh type of multi-speed gear box?
- (e) Define "formative number of teeth" as applied to helical gears and explain its importance in the design of helical gears.
- (f) Explain the theory of hydrodynamic lubrication with neat sketch.
- (g) What is rating life of a rolling contact bearing? Explain it with an example.
- (h) Explain the significance of the bearing characteristic number in the design of sliding contact bearing.
- (i) What are the main design considerations for the parts produced in milling machines?
- (j) Describe about tolerance and surface finish.

PART B

(4 × 10 = 40)

- II. Design a cone clutch to transmit 40 kW at 750 rpm. Also determine the axial force required to transmit the torque and axial force required to engage the clutch. Assume $f = 0.4$ and $p = 0.2 \text{ N/mm}^2$ for friction material ($\alpha = 12.5^\circ$). Take $D_m/b = 6$, where D_m is the mean diameter of friction lining and b is the face width. (10)

OR

- III. Select a V-belt drive to connect a 15 kW, 2880 rpm motor to a centrifugal pump, running at approximately 2400 rpm, for a service of 18 hours per day. The centre distance should be approximately 400 mm. Assume the pitch diameter of driving pulley as 125 mm. (10)

- IV. A 12 kW motor running at 1170 rpm drives a fan through a pair of spur gears (forged steel SAE 1030 pinion and C1 gear) with a reduction ratio of 3:9:1. Design the gear and check for dynamic and wear loads. (10)

OR

- V. The vertical spindle of a drilling machine is to be driven by a pair of straight bevel gears with 20° involute teeth. The speed reduction is 4:1. The drill requires a power of 50 kW at 720 rpm. A service factor of 1.35 may be taken, and choose the steel (C30) as material for both gear and pinion. Design the gear pair. (10)

(P.T.O.)

- VI. It is required to design a main bearing of a 4-stroke oil engine to sustain a load of 6 kN over a shaft of diameter 50 mm. The operating speed of the shaft is 1000 rpm and the operating temperature is 50° C. Determine: (10)
- (i) Dimensions of the bearing
 - (ii) The viscosity of the oil to be used for the bearing and hence suggest appropriate oil
 - (iii) Coefficient of friction
 - (iv) Heat generated
 - (v) Heat dissipated
 - (vi) Heat to be removed by artificial cooling if necessary
 - (vii) Sommerfeld number

OR

- VII. A ball bearing supporting a bevel gear shaft is subjected to a radial load of 4000 N and a thrust load of 5000 N. Select a proper bearing for a speed of 1600 rpm. Life expectancy is 5 years at 10 hours per day. (10)
- VIII. What are the general design recommendations for (i) rolled sections (ii) welded parts? Explain with sketches. (10)

OR

- IX. (a) Explain Indian Standard System of limits and fits. (5)
- (b) What are the general design recommendations for castings? (5)
